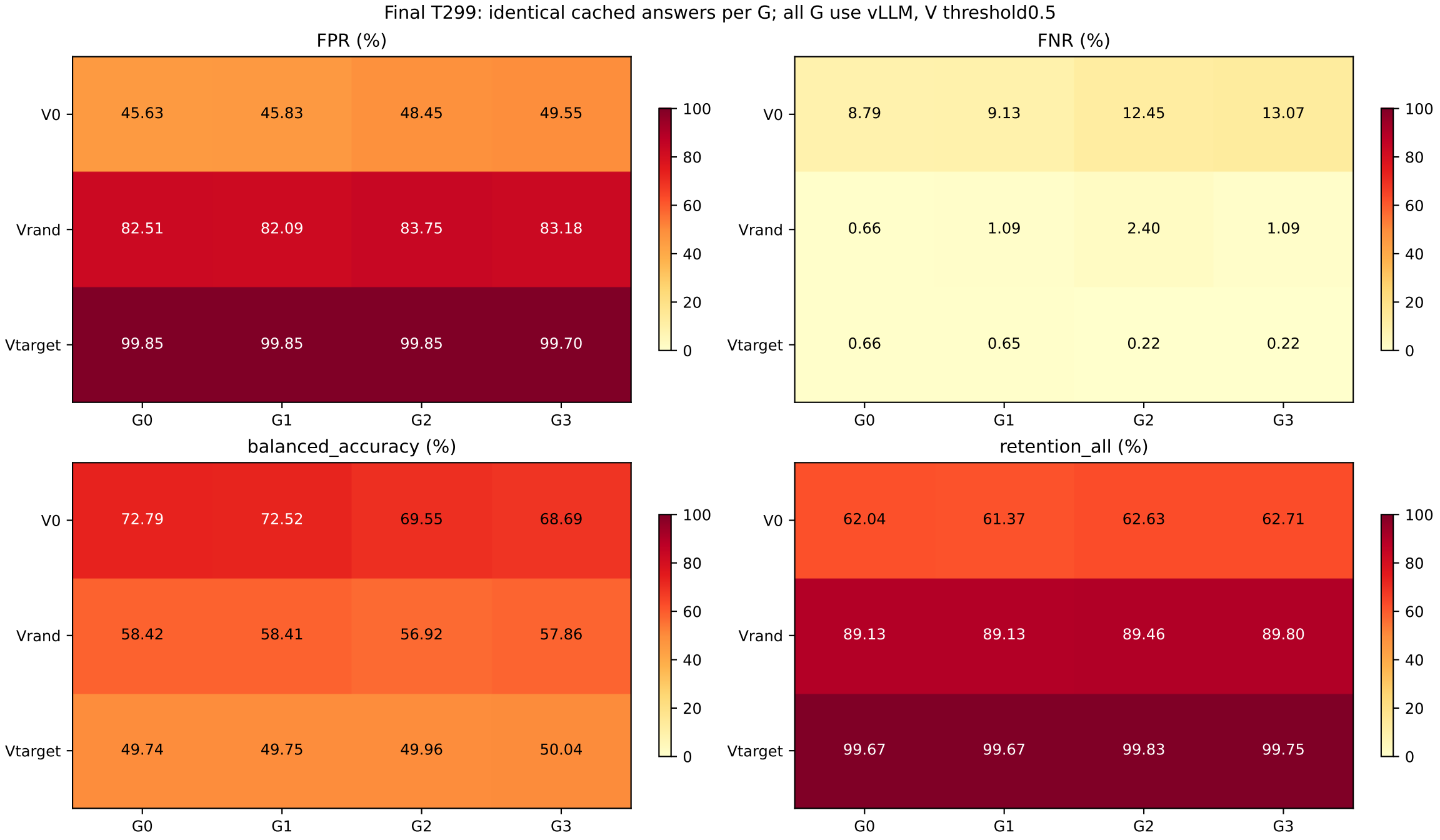
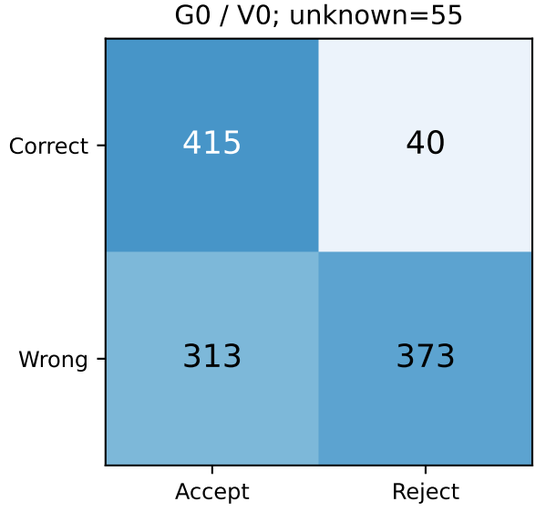
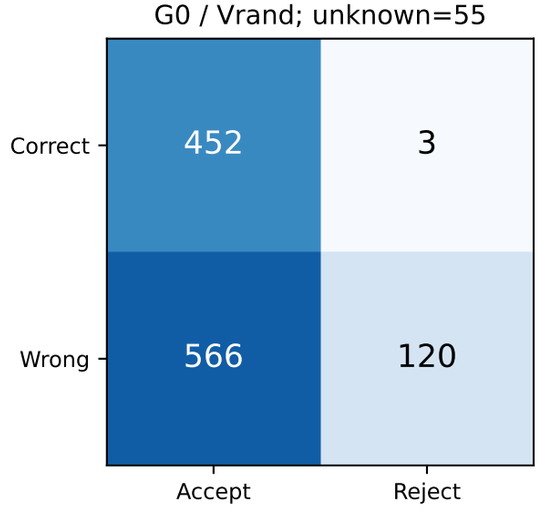
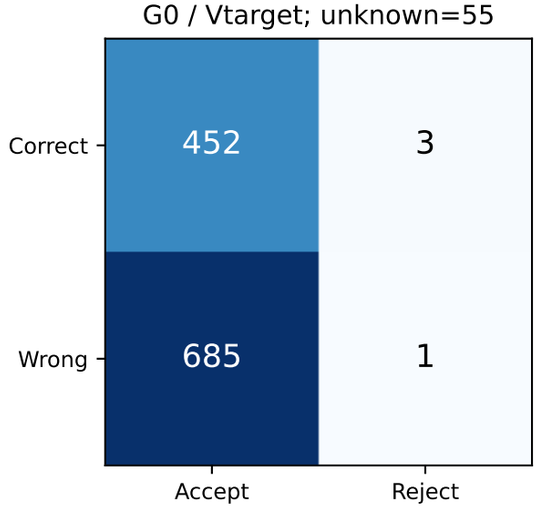
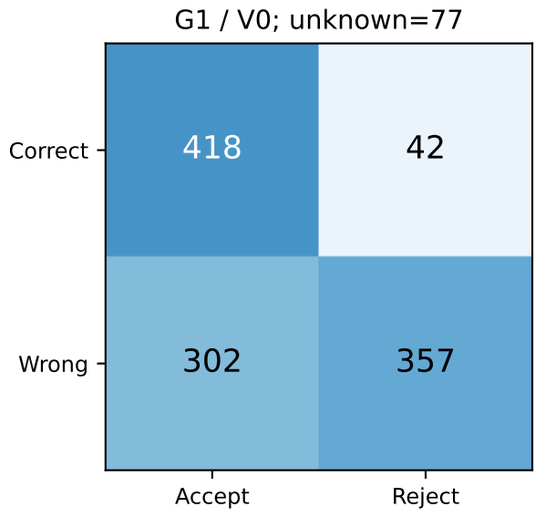

Final T299: identical cached answers per G; all G use vLLM, V threshold0.5
FPR (%)
V0
Vrand
Vtarget
45.63
45.83
48.45
49.55
82.51
82.09
83.75
83.18
99.85
99.85
99.85
99.70
G0
G1
G2
G3
100
80
60
40
20
0
FNR (%)
8.79
9.13
12.45
13.07
0.66
1.09
2.40
1.09
0.66
0.65
0.22
0.22
balanced_accuracy (%)
72.79
72.52
69.55
68.69
58.42
58.41
56.92
57.86
49.74
49.75
49.96
50.04
retention_all (%)
62.04
61.37
62.63
62.71
89.13
89.13
89.46
89.80
99.67
99.67
99.83
99.75

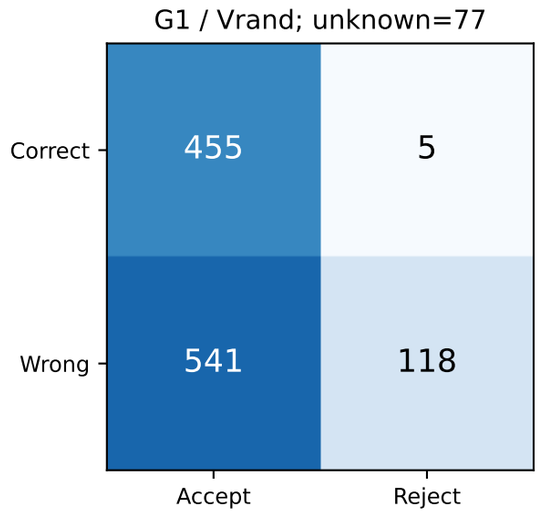
T299 all12 confusion matrices; unknowns excluded, whole invalid question isolated

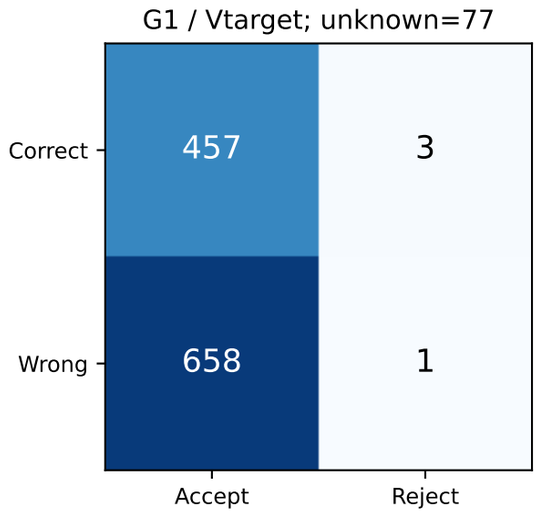


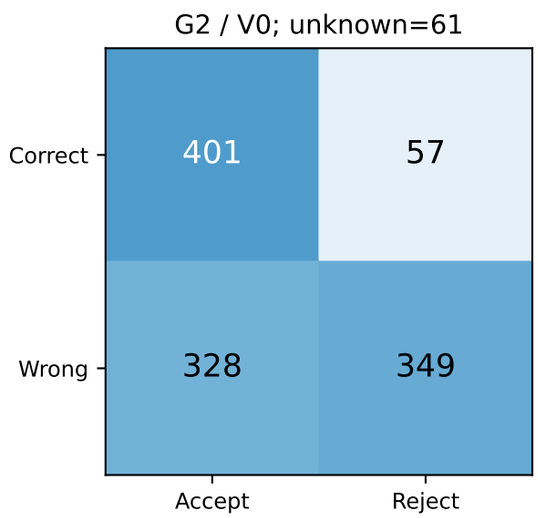


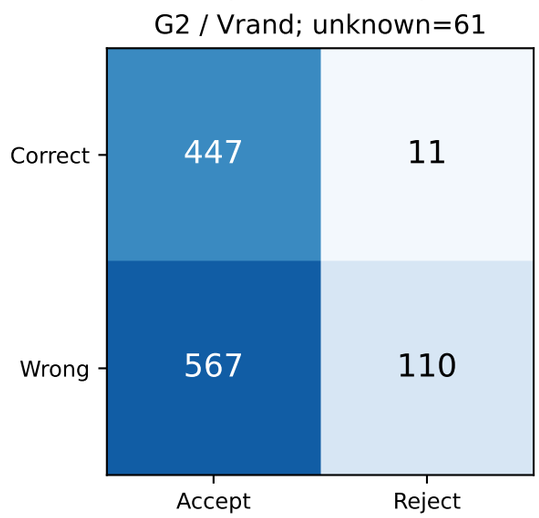


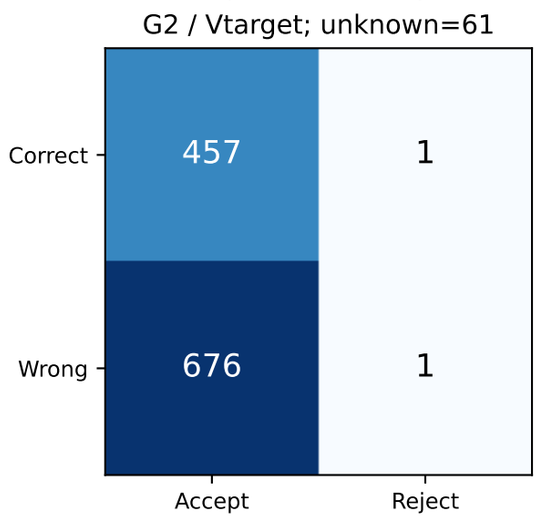


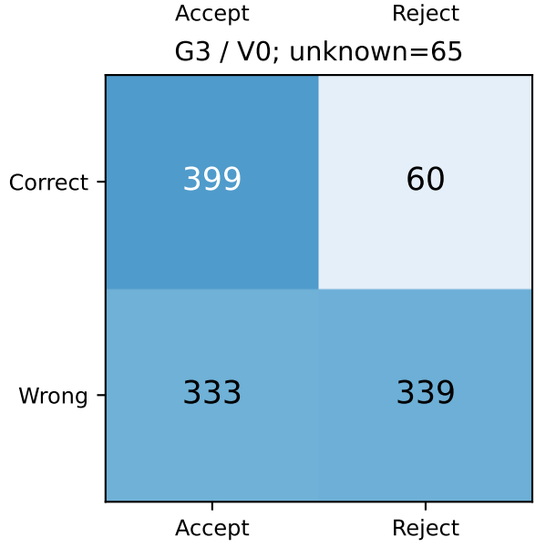


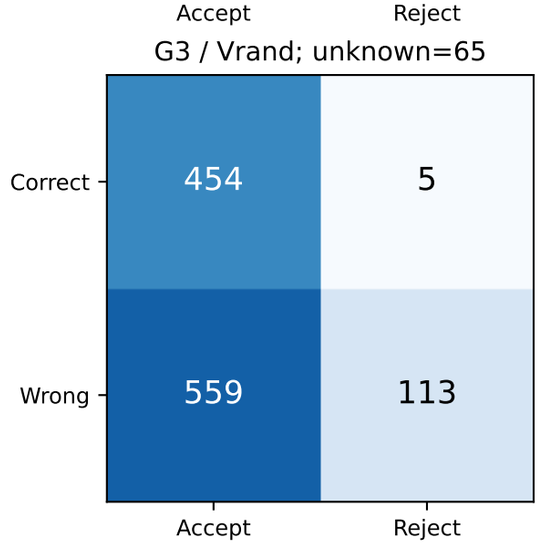


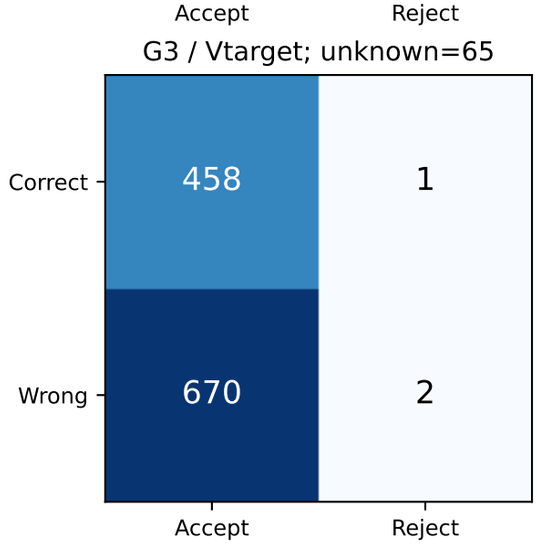


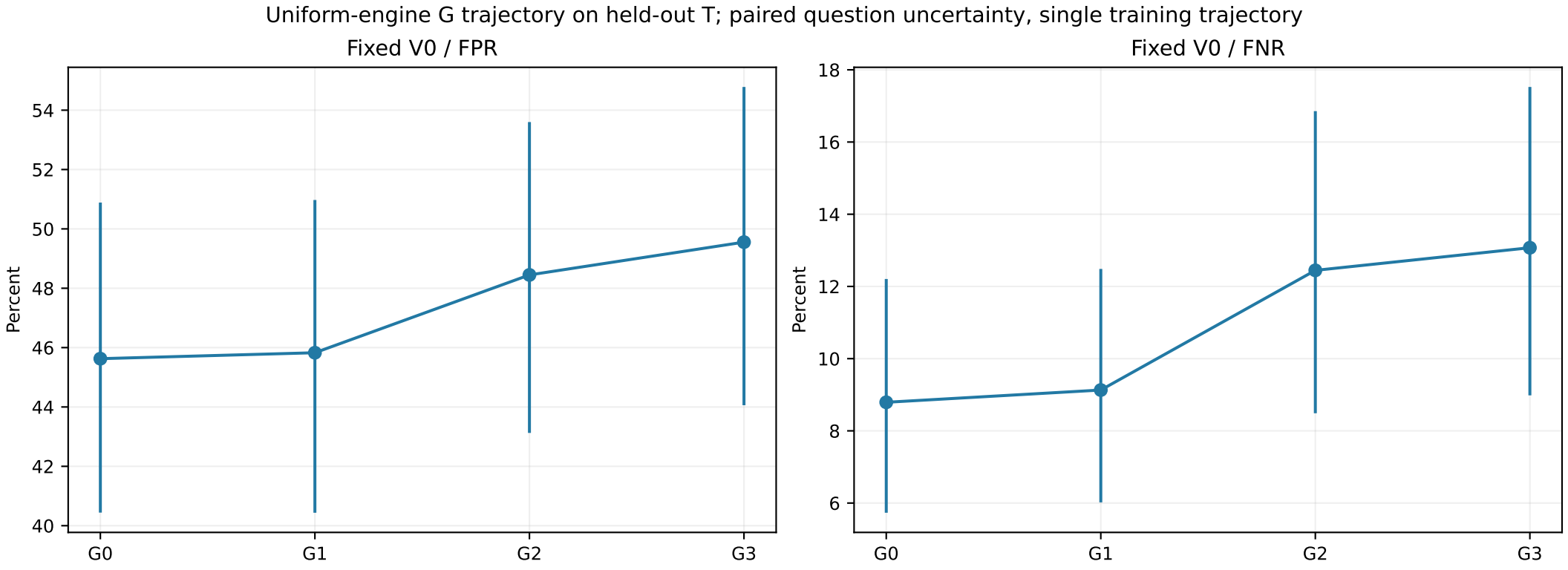


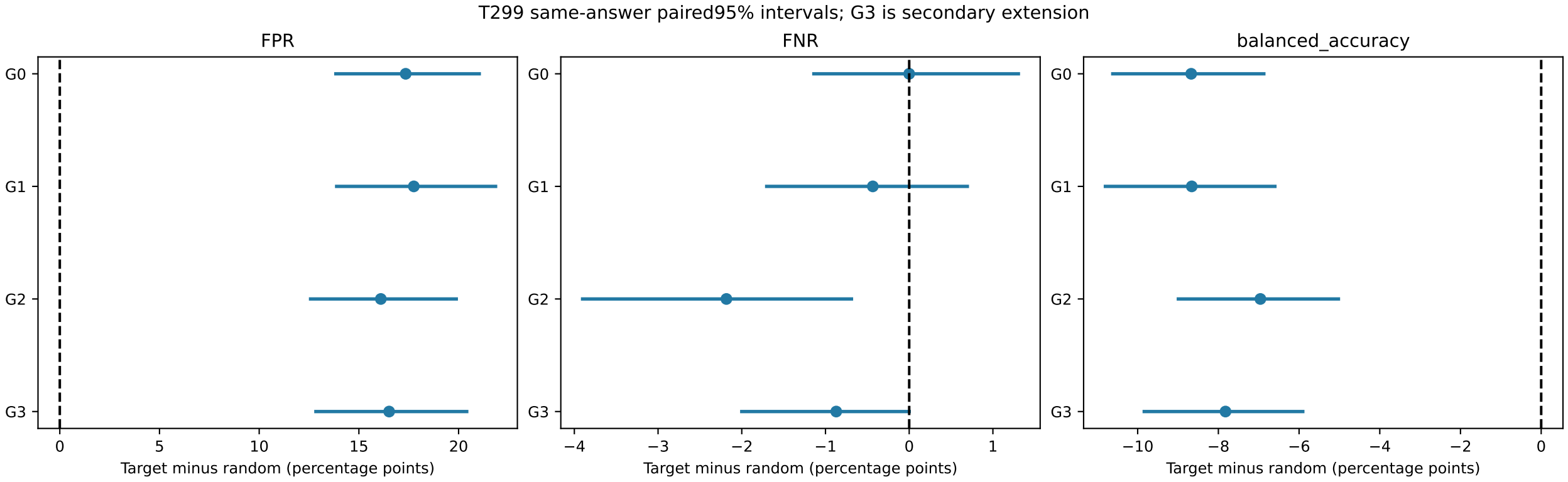


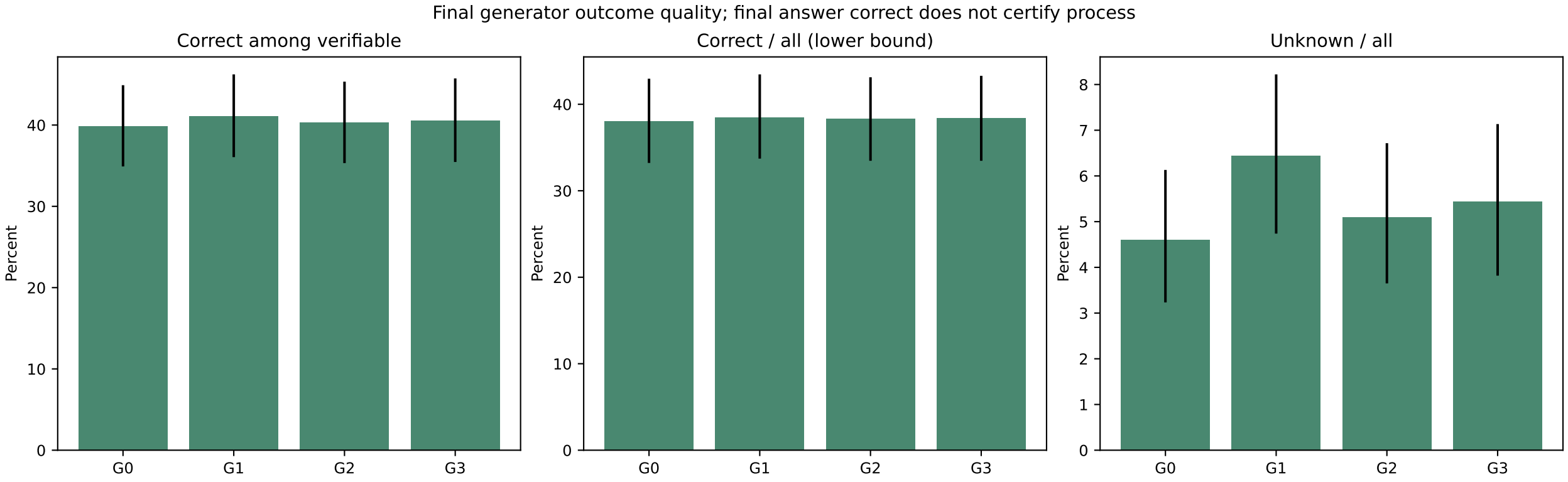


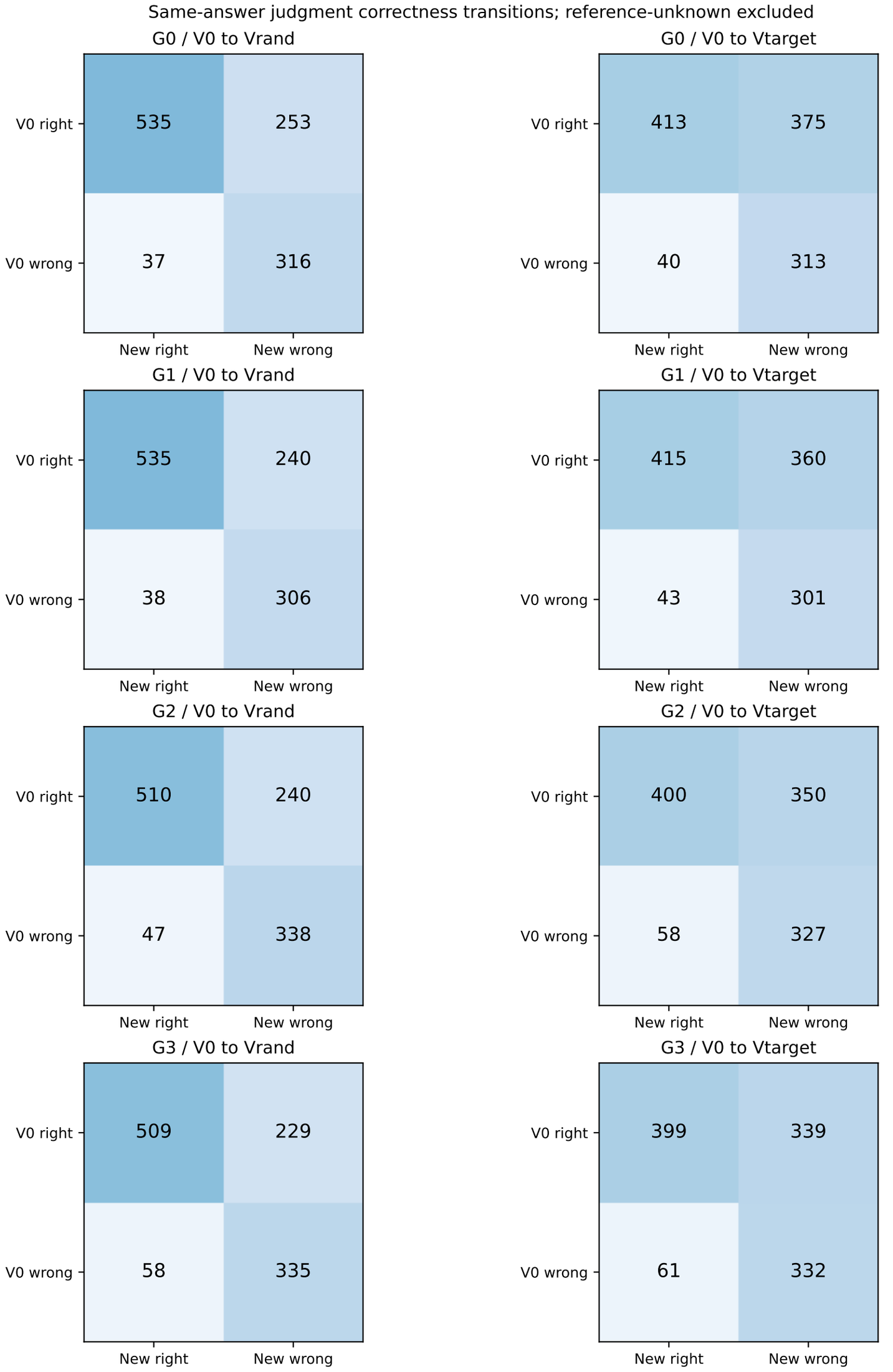





Uniform-engine G trajectory on held-out T; paired question uncertainty, single training trajectory
Fixed V0 / FPR
Fixed V0 / FNR
Percent
54
52
50
48
46
44
42
40
18
16
14
12
10
8
6
G0
G1
G2
G3


T299 same-answer paired95% intervals; G3 is secondary extension
FPR
FNR
balanced_accuracy
G0
G1
G2
G3
0
5
10
15
20
−4
−3
−2
−1
0
1
−10
−8
−6
−4
−2
0
Target minus random (percentage points)
Target minus random (percentage points)
Target minus random (percentage points)


Final generator outcome quality; final answer correct does not certify process
Correct among verifiable
Correct / all (lower bound)
Unknown / all
Percent
0
10
20
30
40
1
2
3
4
5
6
7
8
G0
G1
G2
G3


Same-answer judgment correctness transitions; reference-unknown excluded
G0 / V0 to Vrand
V0 right
V0 wrong
535
253
37
316
New right
New wrong
G0 / V0 to Vtarget
V0 right
V0 wrong
413
375
40
313
New right
New wrong
G1 / V0 to Vrand
V0 right
V0 wrong
535
240
38
306
New right
New wrong
G1 / V0 to Vtarget
V0 right
V0 wrong
415
360
43
301
New right
New wrong
G2 / V0 to Vrand
V0 right
V0 wrong
510
240
47
338
New right
New wrong
G2 / V0 to Vtarget
V0 right
V0 wrong
400
350
58
327
New right
New wrong
G3 / V0 to Vrand
V0 right
V0 wrong
509
229
58
335
New right
New wrong
G3 / V0 to Vtarget
V0 right
V0 wrong
399
339
61
332
New right
New wrong